

MagicChem: A Multi-modal Mixed Reality System Based on Needs Theory for Chemical Education

Tianren Luo
Virtual Reality and Intelligent
Systems Research Institute,
Hangzhou Normal University
Zhipeng Pan
Virtual Reality and Intelligent
Systems Research Institute,
Hangzhou Normal University

Ning Cai
Virtual Reality and Intelligent
Systems Research Institute,
Hangzhou Normal University
Yuze Shen
Virtual Reality and Intelligent
Systems Research Institute,
Hangzhou Normal University

Zheng Li
Virtual Reality and Intelligent
Systems Research Institute,
Hangzhou Normal University
Zhigeng Pan*
Virtual Reality and Intelligent
Systems Research Institute,
Hangzhou Normal University

Jinda Miao
College of Computer
Science and Technology,
Zhejiang University
Mingmin Zhang*
College of Computer
Science and Technology,
Zhejiang University

ABSTRACT

Due to the reasons such as the complicated preparation of reagents, danger, pollution, and lack of teacher resources, there are few chemical experiments that students can actually operate during the chemistry learning, which is not conducive to the chemistry learning and the phenomena principle understanding. Fortunately, MR technology provides us with the possibility of solving the safety issues and the space-time constraints, while the theory of human needs provides us with a way to think about designing a comfortable and stimulant MR system with realistic visual presentation and interaction. This study combines with the theory of human needs to propose a new needs model for virtual experiment. Based on this needs model, we design and develop a comprehensive MR system called MagicChem to verify the needs model. User study shows MagicChem that satisfies the needs model is better than the MR experimental environment that partially meet the needs model. In addition, we explore the application of the needs model in VR environment.

Keywords: Mixed reality, virtual-real occlusion, multi-camera collaboration, virtual-real interaction, chemical education

Index Terms: H.1.2 [Human-centered computing]: Interaction paradigms—Mixed / augmented reality;

1 INTRODUCTION

Maslow's hierarchical needs(MHN) theory believes that human needs are composed of five layers: physiological needs, security needs, belongingness needs, esteem needs, and self-actualization needs. They are the power to inspire and guide individual behavior [3]. The human needs theory is also applicable to the field of experimental teaching, so that students can get a better experience and motivation to obtain a good learning effect.

Guided by needs analysis to design the experiment, the virtual experiment system can make up for the problems of real experiments, and improve the interest and efficiency of students in learning chemical experiments [4]. However, the current research on virtual experimental systems is mainly focused on the single needs point [1] such as experiment simulation, education and teaching, interactive experience and interest. There are few systematic researches on needs model of human-centered virtual experiments, especially in MR experiments.

The theory of human needs provides a new perspective for the system design of virtual experiments. The basic purpose of the virtual experiment system is to simulate the real experiment. However,

*Corresponding author: Zhigeng Pan(e-mail: zgpan@hznu.edu.cn), Mingmin Zhang(e-mail: zhangmm95@zju.edu.cn).

in order for students to obtain better learning effects and interests, it is more important to make the experience of the interactive environment close to the real experiment while having a safe, comfortable, not alone(layers 1-3 in MHN theory) and encouraging (layers 4 and 5 in MHN theory) environment. How to design a robust, comfort and motivating virtual experimental system which can make users have good multi-sensory interactive experience close to the real experiments is the focus of the paper.

2 NEEDS MODEL FOR VIRTUAL EXPERIMENTS

As shown in Figure 1, the paper proposes a human-oriented and systematic needs model for virtual experiments based on the MHN theory.

2.1 Basic Needs

In the first layer of MHN theory, physiological needs are the most basic needs for maintaining human life. But in the virtual experimental environment, food, water and clothing, etc. do not constitute a problem. Therefore, in combination with the application conditions of virtual experiment, we expand the physiological needs to the basic needs, including sensory needs and system needs. Because the sensory experience is the basis of all perceptions of experiments for users and system needs ensure that users use the system effectively.

For sensory needs, we need to simulate the multi-sensory experience of experiments in the MR experimental environment, including visual experience needs, tactile experience needs, action experience needs, listening and speaking experience needs, temperature experience needs and olfactory experience needs. For visual experience needs, spatial presence needs(occlusion consistency) and the presence of virtual objects(illumination consistency) are important in the MR experiments environment. In addition, system needs include basic equipment needs, experimental content needs, usability needs and saving needs.

2.2 Security and Comfort Needs

The second layer of MHN theory is security needs. We extend this layer to security and comfort needs, including safety needs, physical comfort needs and psychological comfort needs.

Security and comfort needs require us to avoid or reduce the psychological and physical pressures(including searching, choosing, tension, worrying, etc.) as much as possible in the interactive process of designing the experimental environment. Such as flames, corrosive liquids and other dangerous objects, we use the form of virtual objects to reduce worries and fears, and use voice interaction to generate virtual devices instead of searching and selecting target devices on the virtual-menu. In addition, we use multi-camera cooperative 3D registration to reduce the physical pressure of users during physical interaction.

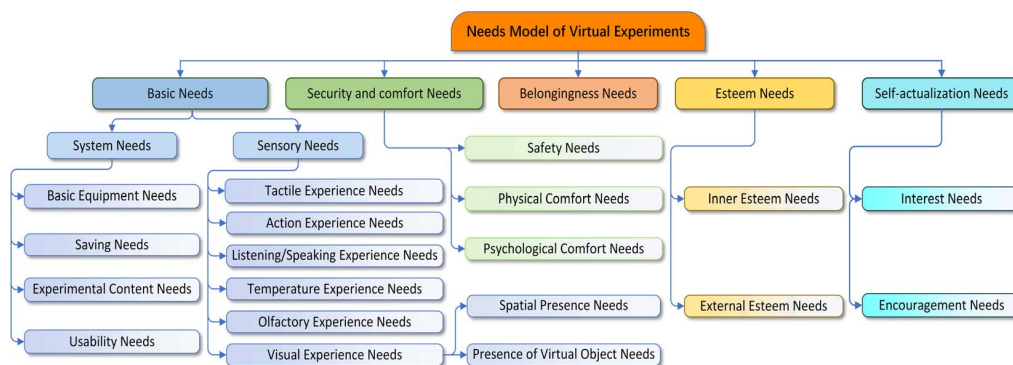


Figure 1: Needs Model of Virtual Experiments

2.3 Belongingness Needs

The third layer of MHN theory is the belongingness needs. The core of this needs is to reduce the loneliness of users. Some studies show that virtual humans can make the user interface rich in expressiveness of emotional interaction, and bring users more sense of participation and enjoyable experience. The paper uses technology of virtual human to create a virtual teacher to meet this needs. In addition, in real chemistry classes, students are accustomed to listening to the teacher's explanation of the principles during the experiment, and they accept this consistent audio-visual learning method. The design allows students to reduce loneliness and enter the state of learning and experiment faster.

2.4 Esteem Needs and Self-actualization Needs

The high-layer needs (layers 4 and 5) of MHN theory are esteem needs and self-actualization needs. The purpose of the two layers' needs is to make users believe in their own power and let users feel the progress they have made step by step through hard learning, so as to achieve the effect of motivation. Therefore, we need to take certain incentive measures in the virtual experiment to stimulate more learning motivation of students. In the paper, the virtual teacher's voice and behavior combine interactive method to meet the two needs.

3 SYSTEM AND EVALUATION

In order to implement the needs model completely, we combined some technologies (Multi-camera cooperative 3D registration, occlusion consistency, multi-modal interaction) of previous work [2], and realized the new functions of illumination consistency, virtual teachers, etc. to develop a system called MagicChem. The interactive user interface of MagicChem is shown in Figure 2.

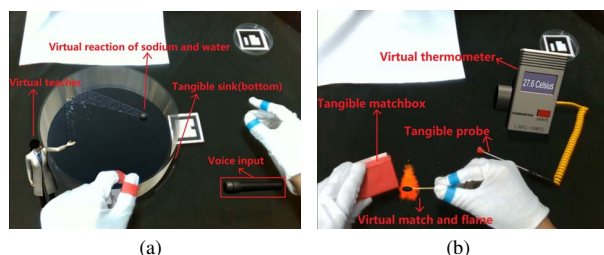


Figure 2: The user interface of MagicChem. (a) Interaction of voice, virtual human and smell. (b) Interaction of tangible, gesture and temperature.

Finally, we developed three ablation groups (MR without consistency of occlusion and illumination, MR with only gesture interaction, VR with HMD) and three typical experimental tasks. We invited 56 users aged 14-16 and divided them into 4 groups to participate in this user study, and use SUS, Nasa-TLX to test the system's experience and impact on users. The results of user study show that, in terms of usability, experience, and learning effects, MagicChem is superior to the two traditional MR environments that partially meet needs models. In terms of application indicators, physical pressure, visual fidelity, interactive naturalness and interest, MagicChem is superior to the VR experiment environment that meets the needs models.

4 CONCLUSION

Combining the human needs theory and the special characteristics of virtual experiments, the paper proposes a human-oriented needs model for virtual experiment and design MagicChem based on this needs model. The results of user study show that the MagicChem is better than the system that does not fully meet the needs model. This also shows the needs model can good guide the design of the virtual experiment system.

ACKNOWLEDGMENTS

The paper is supported by National Key R&D project of China (Grant No. 2018YFB1004902), NSFC project (Grant No. 62077041) and University Student Science and Technology Innovation Activity Plan in Zhejiang Province (Xinmiao Talent Plan, Grant No. 2020R427068).

REFERENCES

- [1] T. Luo, Z. Liu, and Z. Pan. Ar teaching experimental based on multi-camera. In *2018 International Conference on Virtual Reality and Visualization (ICVRV)*, pp. 116–117. IEEE, 2018.
- [2] T. Luo, M. Zhang, Z. Pan, Z. Li, N. Cai, J. Miao, Y. Chen, and M. Xu. Dream-experiment: A mr user interface with natural multi-channel interaction for virtual experiments. *IEEE Transactions on Visualization and Computer Graphics*, 26(12):3524–3534, 2020.
- [3] S. McLeod. Maslow's hierarchy of needs. *Simply psychology*, 1:1–8, 2007.
- [4] S. Ullah, N. Ali, and S. U. Rahman. The effect of procedural guidance on students' skill enhancement in a virtual chemistry laboratory. *Journal of Chemical Education*, 93(12):2018–2025, 2016.